Emotional Multiagent Reinforcement Learning in Spatial Social Dilemmas

Chao Yu, Minjie Zhang, *Senior Member, IEEE,* Fenghui Ren, and Guozhen Tan

***Abstract*—Social dilemmas have attracted extensive interest in the research of multiagent systems in order to study the emergence of cooperative behaviors among selfish agents. Understanding how agents can achieve cooperation in social dilemmas through learning from local experience is a critical problem that has motivated researchers for decades. This paper investigates the possibility of exploiting emotions in agent learning in order to facilitate the emergence of cooperation in social dilemmas. In particular, the spatial version of social dilemmas is considered to study the impact of local interactions on the emergence of cooperation in the whole system. A double-layered emotional multiagent reinforcement learning framework is proposed to endow agents with internal cognitive and emotional capabilities that can drive these agents to learn cooperative behaviors. Experimental results reveal that various network topologies and agent heterogeneities have significant impacts on agent learning behaviors in the proposed framework, and under certain circumstances, high levels of cooperation can be achieved among the agents.**

***Index Terms*—Multiagent learning, social dilemmas, emotions, cooperation.**

I. Introduction

Cooperation is ubiquitous in the real world and can be observed in different organizations ranging from microorganisms and animal groups to human societies [1]. Solving the puzzle of how cooperation emerges among self-interested entities is a challenging issue that has motivated scientists from various disciplines including economics, psychology, sociology and computer science for decades. The emergence of cooperation is often studied in the context of social dilemmas, in which selfish individuals must decide between a socially reciprocal behavior of cooperation to benefit the whole group over time and a self-interested behavior of defection to pursue their own short-term benefits. Social dilemmas often arise in many situations in Multi-Agent Systems (MASs), e.g., file sharing in peer-to-peer (p2p) systems, load balancing/packet routing in wireless sensor networks and bandwidth allocation/frequency detection in telecommunication systems [2]. For this reason, mechanisms that promote the emergence of cooperation in social dilemmas are of great interest to researchers in MASs.

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C. Yu and G. Tan are with the School of Computer Science and Technology, Dalian University of Technology, Dalian 116024, China (e-mail: {cy496,gztan}@dlut.edu.cn).

M. Zhang and F. Ren are with the School of Computer Science and Software Engineering, University of Wollongong, Wollongong, NSW 2522, Australia (e-mail:{minjie,fren}@uow.edu.au).

In the literature, a variety of metaphors can be adopted to study social dilemmas among self-interested agents, in which Prisoners' Dilemma (PD) is the most widely used one. In PD, each player has two actions: 'cooperate' (C), which is a socially reciprocal behavior and 'defect' (D), which is a self-interested behavior. Consider the typical pay-off matrix of the PD game in Table I. The rational action for both agents is to select D because choosing D ensures a higher pay-off for either agent no matter what the opponent does. In other words, mutual defection (DD) is the unique Nash equilibrium that both agents have no incentive to deviate from. However, DD is not the Pareto optimal because, if both agents select C, each agent would be better off (i.e., $H > P$) and both agents together would receive a higher social reward (i.e., $2H > T + S$). When played repeatedly, which is called the Iterated Prisoner's Dilemma (IPD), it may be beneficial for an agent to cooperate in some rounds, even if this agent is selfish, in the hope of a reciprocal cooperative behavior in the long run. A social dilemma then arises for the agent to decide whether to cooperate for the long-term benefits or to defect for the selfish short-term benefits.

TABLE I
THE PAY-OFF MATRIX FOR THE PD GAME (PAY-OFF OF THE ROW PLAYER IS SHOWN FIRST; T IS THE TEMPTATION TO DEFECT, H IS THE REWARD FOR MUTUAL COOPERATION, P IS THE PUNISHMENT FOR MUTUAL DEFECTION AND S IS THE SUCKER'S PAY-OFF. TO BE DEFINED AS A PD GAME, THE FOLLOWING CONSTRAINS SHOULD BE MET: $T > H > P > S$ AND $2H > T + S$.)

	Cooperate (C)	Defect (D)
Cooperate (C)	H(= 3),H(= 3)	S(= 0),T(= 5)
Defect (D)	T(= 5),S(= 0)	P(= 1),P(= 1)

Until now, various mechanisms, such as kin selection [3] and social diversity [4], have been proposed to explain the emergence of cooperation in social dilemmas. Most of these mechanisms, however, are based on the Evolutionary Game Theory (EGT) [5], [6], with a focus either on the macro-level population dynamics using replicator functions or on the agent-level strategy dynamics based on predefined imitation rules. Real animals and humans, however, not only replicate or mimic others, but can learn efficient strategies from past interaction experience. In fact, this experience-based learning capability is important in building intelligent agents that can align human behavior, particularly when designers cannot anticipate all situations that the agents might encounter.

A major family of experience-based learning is Reinforcement Learning(RL) [7], which enables an agent to learn an optimal strategy through trial-and-error interactions with an environment. When multiple agents conduct their learning at

the same time, which is called Multi-Agent Reinforcement Learning (MARL) [8], each agent faces a non-stationary environment and therefore each agent's individually optimal strategy does not necessarily guarantee a globally optimal performance for the whole system. In fact, in the setting of IPD, directly applying distributed MARL approaches with no additional mechanisms implemented will end up with convergence to the Nash equilibrium of mutual defection [9]. The convergence to the Nash equilibrium occurs because both agents adopt the best-response actions during learning. As a result, neither agent can achieve a dominant position by choosing defection to exploit its opponent because the opponent will eliminate such dominance by also choosing defection, resulting in mutual defection between the agents.

This paper investigates the possibility of exploiting emotions to modify agent learning behaviors in MARL in order to facilitate cooperation in social dilemmas. This research is driven by the fact that emotions can be an important heuristic to assist agents' decision making during learning [10], [11]. Emotions can play a fundamental role in learning by eliciting physiological signals that bias agents' behaviors toward maximizing reward and minimizing punishment. Therefore, MARL mechanisms, in essence, should rely on some emotional cues to indicate the advantage or disadvantage of an event.

In particular, this paper focuses on spatial versions of social dilemmas by considering topological structures among agents in order to study the impact of local interactions on the emergence of cooperation. A double-layered emotional MARL framework is proposed to endow agents with internal cognitive and emotional capabilities that can drive these agents to learn cooperative behaviors in social dilemmas. In the framework, two fundamental variables, individual wellbeing and social fairness, are considered in the appraisal of emotions. Different structural relationships between these two appraisal variables, represented by different emotion derivation functions, can derive various intrinsic emotional rewards for agent learning. In the meta-level inner-layer learning, two emotion derivation functions compete with each other in order to dominate the agent's emotion derivation process, while in the outer-layer learning, an explicit strategy of behaviors can be learnt based on the selected emotion derivation function. Experimental results reveal that different ways of appraising emotions, various network topologies and agent heterogeneities have significant impacts on agent learning behaviors, and high levels of cooperation can be achieved under certain circumstances.

This paper advances the state of the art in two aspects:

1) **From the theoretical perspective**, this paper provides a psychological explanation of cooperative behaviors in human societies. Psychological scientist have shown that emotions can play a vital role for human beings to successfully solve social dilemmas [12]. Research in the psychology paradigm, however, is mainly based on empirical experiments involving human participants, and therefore, cannot provide an explicit interpretation of the observed cooperative behaviors. This paper conquers this limitation by clearly quantifying the emotion components and modeling the emotional reasoning process so that we can have a deeper understanding of the psychological mechanisms behind human cooperative behaviors in social dilemmas.
2) **From the technical perspective**, this paper provides a computationally sound model that incorporates emotions into MARL in order to solve social dilemmas. In the current literature, emotion-based mechanisms have been widely applied in agent learning [13], [11], [14], with an aim to facilitate learning efficiency or to adapt a single agent to dynamic and complex environments. On the other hand, several studies [15], [16] have examined social dilemmas based on rule-based emotional frameworks, in which agents should decide whether to cooperate directly based on the emotions derived from a predefined rule system. This paper bridges these two research directions by incorporating emotions into MARL in order to facilitate the emergence of cooperation in social dilemmas.

The remainder of the paper is organized as follows. Section II introduces spatial social dilemmas and MARL. Section III describes the proposed framework. Section IV illustrates the implementation of the emotional MARL framework in social dilemmas. Section V shows the experimental studies. Section VI discusses some related work. Finally, Section VII concludes this paper with directions for future research.

II. Spatial Social Dilemmas and MARL

This section introduces spatial social dilemmas and some fundamental concepts in MARL.

A. Spatial Social Dilemmas

A spatial social dilemma is defined by Definition 1 [17].

Definition 1: A **spatial social dilemma** can be represented as a graph $G = (V, E)$, where $V = \{v_1, ..., v_n\}$ is a set of vertices (agents) and $E \subseteq V \times V$ represents a set of edges, each of which connects two interacting vertices (agents) playing a social dilemma game.

Definition 2: Given a spatial social dilemma G, the **neighbors** of agent i, which are denoted as $N(i)$, are a set of agents such that $N(i) = \{v_j \mid (v_i, v_j) \in E\}$ and $N(i) \subset V$.

This paper focuses on the following three types of networks to represent a spatial spatial social dilemma.

(1) **Grid networks**. A grid network is a two-dimensional lattice with four neighbors for each inner node, three neighbors for each boundary node, and two neighbors for each corner node. In reality, parallel computing clusters and multi-core processors are usually organized as a grid network. Let GR_N denote a grid network, where N is the number of nodes.

(2) **Small-world networks**. This kind of network represents the small-world phenomenon in many natural, social and computer networks, where each node has only a small number of neighbors and yet can reach any other node in a small number of hops. Typical examples include collaboration networks of film actors or academic researchers, and friendship networks of high school students [18]. Small-world networks feature a high clustering coefficient and a short average path length. Let $SW_N^{k,\rho}$ denote a small-world network, where k is the average

size of the neighborhood, and $\rho \in [0, 1]$ indicates the different orders of network randomness.

(3) **Scale-free networks**. This kind of network is characterized by the power law of degree distribution of nodes, which means that a few "rich" nodes have high connectivity degrees, while the remaining nodes have low connectivity degrees. The probability that a node has k neighbors is roughly proportional to $k^{-\lambda}$. Examples of scale-free networks include the network of citations of scientific papers and links between web pages on the World Wide Web [18]. These networks exhibit the feature of "preferential attachment", which means that the likelihood of connecting to a node depends on the connectivity degree of this node. Let $SF_N^{k,\lambda}$ denote a scale-free network, where k and λ are coefficients in the probability $k^{-\lambda}$, and N is the number of nodes.

B. MARL

MARL is a multiple agent extension of single agent RL. In RL, the environment of an agent is described by a Markov decision process. A *Markov Decision Process* (MDP) can be formally defined by a 4-tuple $M = (S, A, P, R)$, where S is a finite state space, A is a set of actions, $P(s, a, s') : S \times A \times S \to [0, 1]$ is a Markovian transition function when the agent transits from state s to s' after taking action a and $R : S \times A \to R$ is a reward function that returns the immediate reward $R(s, a)$ to the agent after taking action a in state s. An agent's *policy* $\pi : S \times A \to [0, 1]$ is a probability distribution that maps a state $s \in S$ to an action $a \in A$ of the agent.

The goal of an agent in MDP is to learn a policy π so as to maximize the expected discounted reward $Q^\pi(s, a)$ for each state $s \in S$ and $a \in A$, which is given by Equation 1,

$$Q^\pi(s, a) = E_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | s_0 = s, a_0 = a \right] \quad (1)$$

where E_π is the expectation of policy π, and $\gamma \in [0, 1)$ is a discount factor.

For any finite MDP, there is at least one *optimal policy* π^*, such that $Q^{\pi^*}(s, a) \geq Q^\pi(s, a)$ for every state $s \in S$ and action $a \in A$. The optimal policy π^* can be computed by using linear programming or dynamic programming techniques if an agent fully knows the reward and transition functions of the environment. When these functions are unknown, finding an optimal policy in MDP can be solved by RL methods in which an agent learns through trial-and-error interactions with its environment [7]. One of the most important and widely used RL approach is Q-learning [19], which is an off-policy model-free temporal difference control algorithm. In Q-learning, an agent makes a decision through the estimation of a set of Q values. The one step updating rule of Q values is given by Equation 2, where $\alpha \in (0, 1]$ is the learning rate.

$$Q_{t+1}(s, a) = Q_t(s, a) + \alpha_t [R(s, a) + \gamma \max_{a'} Q_t(s', a') - Q_t(s, a)] \quad (2)$$

In MARL, each agent takes an action on the environment, and the state transitions and rewards are the result of the joint action of all agents. A straightforward way to solve MARL problems is to let agents learn based on the complete information from all other agents. This kind of centralized approach, however, becomes infeasible quickly because the search space grows rapidly with the complexity of agent behaviors, the number of agents and the size of domains. Moreover, in many practical situations, agents might not be able to have access to the complete information from other agents due to observability limits and communication constraints.

Another straightforward way to solve MARL problems is to let each agent learn individually using its own local state/action and individual reward information so that the curse of dimensionality using a centralized learning approach can be avoided. The main challenge using this fully distributed learning approach, however, is that each learner must adapt its behavior in the context of other co-learners so that the learning environment becomes non-stationary for each learner. Dynamics in such a non-stationary environment can cause the learning goal of a learner to change continuously, making MARL into a "moving-target learning" problem [8]. Therefore, the guarantee of optimal convergence in single agent RL no longer holds in general cases of MARL.

III. The Emotional MARL Framework

Applying distributed MARL approaches to solve social dilemmas is a very challenging issue because agents not only need to learn the structure of the interaction game, but also need to learn how to play the game optimally in a non-stationary environment. Various mechanisms have been proposed to boost cooperation by altering the definition of local state and individual reward in MARL. For example, previous studies [20], [21] defined local state as the sensations of past actions of the interacting agents, while other studies [22], [23], [24] evolved or reshaped individual rewards during interactions so that agents could rapidly achieve cooperation.

Following the second paradigm, we proposed an emotional MARL framework [25] to emotionally derive the agent's individual reward in order to facilitate the emergence of cooperation among the agents. In the framework, an agent takes an action on the external environment, observes its neighbors' actions and receives extrinsic rewards from the environment. The internal environment, which includes an emotion appraisal model and an emotion derivation model, generates intrinsic emotional rewards that are used as reinforcement signals to adapt learning behaviors. The framework in [25], however, only considers a single emotional cue during learning, which contradicts real-life situations when humans usually make decisions based on multiple (sometimes conflicting) emotional cues. To better model this phenomenon, we propose an extended emotional learning framework in Fig. 1.

The emotion appraisal model transforms an agent's belief about its relationship with the environment into a set of quantified appraisal variables. The framework uses two fundamental variables to appraise emotions by considering not only an agent's individual wellbeing but also its sense of social fairness. These two variables are considered because it is believed that emotions could be generally considered to be composed of interacting elementary variables from two main

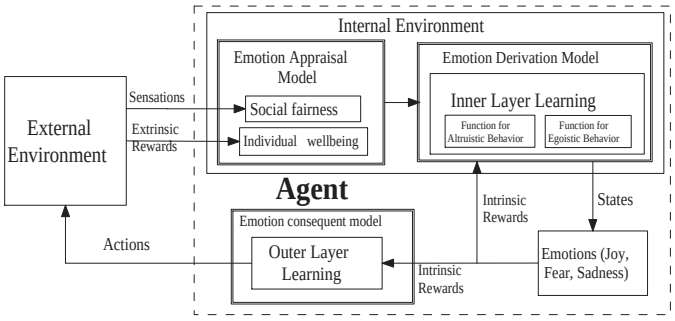


Fig. 1. The proposed framework (adapted from [25]).

dimensions: the motivational dimension dealing with individual goals, needs and pleasantness, and the social dimension dealing with social norms, justice and fairness [26].

The emotion derivation model maps appraisal variables to an emotional state, and specifies how an agent reacts emotionally once a pattern of appraisals has been determined. By combining the variables using different emotion derivation functions, different internal cues can be defined for the agents to emotionally react to the environment. In the framework, each agent has two contradictory emotion derivation functions. One indicates an altruistic behavior and the other indicates a selfish egoistic behavior. Both functions map the appraisal variables to an emotion state (i.e., intrinsic emotional reward) based on agents' different evaluations of the structural relationship between appraisal variables and specific emotions. The intrinsic reward is then used to update the strategy of choosing the emotion derivation functions. This learning process, referred to as "inner-layer learning", enables contradictory emotional cues to compete with each other during interactions so that a certain emotional cue could emerge as a dominant factor in the agent's emotion derivation process.

The emotion consequent model maps emotions into some behavioral or cognitive changes. In the framework, the intrinsic emotional rewards generated from the internal environment are used directly as reinforcement signals to bias the agent's learning behaviors. This process is referred to as "outer-layer learning", as opposed to the inner-layer learning process used to adapt an agent's internal cues of deriving emotions.

IV. Emotional MARL in Spatial Social Dilemmas

This section presents the implementation of the proposed framework in spatial social dilemmas.

A. Appraisal of Emotions

This subsection restates the appraisal of emotions in spatial social dilemmas proposed in [25].

1) Appraisal of Social Fairness: In order to evaluate the social fairness, an agent needs to assess its own situation in the environment represented by its neighborhood context C_n [25]:

$$C_n = \frac{1}{N}\sum_{i=1}^{i=N}(\frac{n_c^i - n_d^i}{M}), \tag{3}$$

where n_c^i and n_d^i are the numbers of C action and D action adopted by neighbor i in M steps[1], respectively, and N is the number of neighbors of the focal agent.

In Equation 3, $\frac{n_c^i - n_d^i}{M}$ indicates the cooperativeness of neighbor i. If neighbor i cooperates more often ($n_c^i - n_d^i \geq 0$), it is considered to be more cooperative. Otherwise, it is considered to be more defective. A focal agent's neighborhood context $C_n \in [-1, 1]$, thus, indicates the extent of cooperativeness of the environment, with $C_n = 1$ indicating a fully cooperative environment, $C_n = -1$ indicating a fully uncooperative environment, and $C_n = 0$ indicating a neutral environment.

An agent's sense of fairness is then to evaluate its own situation in such a neighborhood context [25].

$$F = C_n \times \frac{n_c - n_d}{M}, \tag{4}$$

where $F \in [-1, 1]$ represents the agent's sense of social fairness, n_c is the number of C action and n_d is the number of D action adopted by the agent in M steps.

From Equation 4, we can see that when the environment is cooperative ($C_n > 0$), an agent senses fairness if it cooperates more often ($n_c > n_d$) and unfair if it defects more often ($n_c < n_d$). On the contrary, when the environment is defective ($C_n < 0$), an agent senses fairness if it defects more often ($n_c < n_d$) and unfair if it cooperates more often ($n_c > n_d$).

In economic and behavioral sciences, fairness can be defined variously. The most widely used definition is based on the inequity aversion function [27], which indicates that people resist inequitable outcomes by giving up some material payoff to move in the direction of more equitable outcomes. The use of inequity aversion function relies on the comparison of an agent's own utility with its counterpart's utility. The fairness defined by Equation 4, however, is simply based on an agent's evaluation of its own situation in a dynamic neighborhood context. The decision making associated with the appraisal of fairness is then formulated in the emotion derivation process, which will be described in detail in Subsection IV-B.

2) Appraisal of Individual Wellbeing: Agents also need to care about their own individual wellbeing in terms of maximizing utilities. Three different approaches are proposed in [25] to appraise an agent's individual wellbeing as follows.

- **Absolute value-based approach**: A straightforward way to appraise an agent's individual wellbeing is to use its absolute wealth as an evaluation criterion [25].

$$W = \frac{2R_t - M \times (T - S)}{M \times (T - S)}, \tag{5}$$

  where R_t is the accumulated reward in M interaction steps at episode t and $M \times (T - S)$ is the normalization factor to confine W to $[-1, 1]$, with $W = 1$ indicating the highest wellbeing of the agent and $W = -1$ indicating the lowest wellbeing of the agent.
- **Variance-based approach**: It is argued that low wealth does not necessarily imply a negative reaction and high wealth does not necessarily imply a positive reaction of

[1]To model the memory of agents, a learning episode consists of several interaction steps, which means that the learning information will be updated only at the end of M interaction steps.

the agents [13]. Therefore, individual wellbeing can be defined as positive and negative variations of an agent's absolute wealth, which is given by Equation 6 [25].

$$W = \frac{R_{t+1} - R_t}{M \times (T - S)}, \tag{6}$$

where R_t and R_{t+1} are the accumulated reward collected during M interaction steps at learning episode t and the following episode $t+1$, respectively, and $R_{t+1} - R_t$ is the variation of an agent's absolute wealth.

- **Aspiration-based approach**: The aspiration-based approach appraises the agent's individual wellbeing by comparing the reward achieved with an adaptive aspiration level, which is given by Equation 7 [25].

$$W = \tanh[h(\frac{R_t}{M} - A_t)], \tag{7}$$

where $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is the monotonically increasing hyperbolic tangent function that maps variable $x \in \Re$ into $[-1, 1]$, $h > 0$ is a scalable parameter, and A_t is the aspiration level, which can be updated by:

$$A_{t+1} = (1 - \beta)A_t + \beta\frac{R_t}{M}, \tag{8}$$

where $\beta \in [0, 1]$ is a learning rate and $A_0 = \frac{R+T+S+P}{4}$ is the initial aspiration level.

The basic idea of the aspiration-based approach is that an agent feels positive about its own wellbeing state ($W \geq 0$) if the received reward is higher than its internal aspiration level ($\frac{R_t}{M} \geq A_t$), and feels negative about its own wellbeing state ($W < 0$) if the received reward is lower than its internal aspiration level ($\frac{R_t}{M} < A_t$). The aspiration level is then updated as a weighted average between the received reward and the current aspiration level, which is given by Equation 8.

B. Derivation of Emotions

After appraising the two fundamental variables in the appraisal of emotions, the emotion derivation model then maps these variables to an emotion state through an emotion derivation function. The emotion derivation function indicates the structural relationship between appraisal variables and specific emotion states, and therefore stipulates how an agent reacts emotionally once a pattern of appraisals has been determined.

*1) **Formulation of Emotion Derivation Functions**:* In [25], a structural model of emotion appraisal was proposed to explain the relation between appraisals and the emotions they elicit. In this model, based on the differentiation of the emotion appraisal process [28], an appraisal variable can be defined as a core appraisal variable (denoted as c) or a secondary appraisal variable (denoted as s). The core appraisal variable determines the desirability of an emotion through the agent's evaluation of its situation, while the secondary appraisal variable indicates the intensity of such an emotion based on the agent's evaluation of its coping ability. An emotion derivation function of emotion x, F_x, therefore, can be formally defined as follows:

Definition 3: An ***emotion derivation function*** F_x can be defined as $E_x(c, s) \overset{F_x}{=} f(D_x) \cdot g(I_x)$, where $0 \leq E_x(c, s) \leq 1$ is the overall state of emotion x, $0 \leq f(D_x) \leq 1$ is the core derivation function that increases monotonically with the desirability of emotion x (i.e., $0 \leq D_x \leq 1$), and $0 \leq g(I_x) \leq 1$ is the secondary derivation function that increases monotonically with the intensity of emotion x (i.e., $-1 \leq I_x \leq 1$).

Two emotion derivation functions can be defined as follows.

Definition 4: An agent that derives its emotions using the ***Fairness-Wellbeing (FW)*** emotion derivation function puts social fairness F as the core appraisal variable (i.e., $c \leftarrow F$) and then derives its emotions based on the increase or decrease in its own wellbeing W (i.e., $s \leftarrow W$).

Definition 5: An agent that derives its emotions using the ***Wellbeing-Fairness (WF)*** emotion derivation function puts its own wellbeing W as the core derivation variable (i.e., $c \leftarrow W$) and then derives its emotions based on its sense of social fairness F (i.e., $s \leftarrow F$).

An emotion can be a positive one or negative one, depending on whether the encounter is consistent with the agent's motivational goals. An emotion is positive if it is joy and negative if it is $fear$, $sadness$ or $anger$. These four emotions are modeled as they are believed to be the most fundamental emotions in humans [10], [13]. The final state of emotion x is then used as the intrinsic reward R_{int} for agent learning:

$$R_{int} = \begin{cases} E_x & \text{if emotion } x \text{ is positive,} \\ -E_x & \text{if emotion } x \text{ is negative.} \end{cases} \tag{9}$$

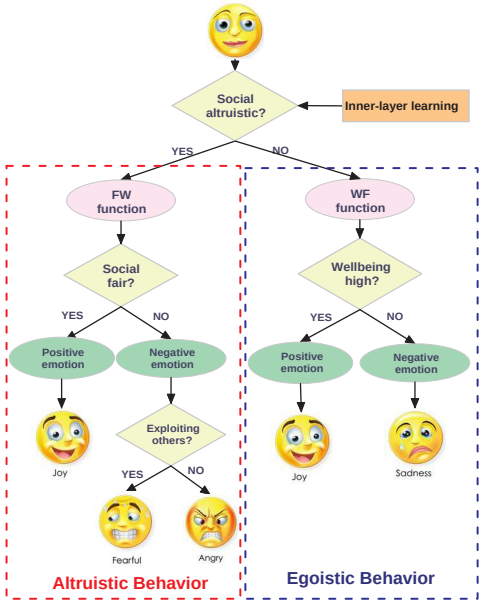


Fig. 2. The reasoning process of deriving emotions.

Fig. 2 plots an agent's reasoning process of deriving emotions. An agent determines to behave in a social altruistic way or in a selfish egoistic way according to the strategy in the inner-layer learning. The inner-layer learning process can be carried out using general MARL approaches. If the agent behaves as an altruist, it derives its emotions based on the FW function, and checks whether the environment is fair. The

agent will sense positive emotion of joy if the environment is fair, and negative emotions of $fear$ or $anger$ according to whether it is exploiting others or not. If the agent behaves as an egoist, it derives its emotions based on the WF function, and checks whether its individual wellbeing is high. The agent will sense positive emotion of joy if its individual wellbeing is high, and negative emotion of $sadness$ otherwise.

- **Altruistic behavior based on the FW function**: An agent using the FW function to derive its emotions is a socially aware agent that pays more attention to social fairness than to its individual wellbeing in appraising its emotions. As the first priority of an FW agent is to pursue social fairness, the agent will be in a positive state of joy in a fair environment (i.e., $F > 0$), because the situation is consistent with the agent's motivational goal. Desirability D_{joy} is then equivalent to the value of core derivation variable of F. The intensity of emotion joy is then based on the increase or decrease of the agent's wellbeing. An increase of the individual wellbeing in a fair environment indicates that the socially aware agent can achieve selfish interest while at the same time pursue its core motivational goal of social fairness. The final state of the emotion of joy, therefore, can be calculated as $E_{joy}(F, W) = f(F) \cdot g(W)$.
  On the contrary, the agent will feel negative in an unfair environment (i.e., $F < 0$) because the situation is inconsistent with the agent's goal of pursuing social fairness. The unfair environment can be caused by two situations: the agent defects more often in a cooperative environment or the agent cooperates more often in a defective environment. The socially aware agent will feel $fear$ in the former situation because it is exploiting its neighbors by choosing defection [13], while in the latter situation, the agent is in emotion of $anger$ because it is being exploited by its neighbors [15]. In both situations, desirability of the negative emotion is equivalent to the value of $-F$. If the agent feels $fear$, the intensity of emotion $fear$ then increases monotonously with the wellbeing of the agent ($I_{fear} = W$) because the socially aware agent realizes that it is exploiting its neighbors for an increase in its own wellbeing. The higher the wellbeing, the higher intensity of fear of the agent. The final state of the emotion of $fear$ can be then calculated as $E_{fear}(F, W) = f(-F) \cdot g(W)$. In contrast, if the agent feels angry, the intensity of emotion $anger$ then increases inversely with the wellbeing state of the agent ($I_{anger} = -W$) because a lower wellbeing will result in a higher intensity of emotion $anger$. The final state of emotion $anger$ can be then calculated as $E_{anger}(F, W) = f(-F) \cdot g(-W)$.
- **Egoistic behavior based on the WF function**: An agent using the WF function to derive its emotions is more like an egoist that pays more attention to its own wellbeing than to social fairness to determine its emotions. As the first priority of a selfish WF agent is to pursue its own benefits, the agent will be in a positive state of joy when the agent is in a high wellbeing state (i.e., $W > 0$) because the situation is consistent with the agent's motivational goal. The desirability of emotion joy, D_{joy}, is then equivalent to the value of core derivation variable W. The intensity of emotion joy is then based on the agent's sense of social fairness. A high social fairness associated with a high individual wellbeing indicates that the selfish agent can achieve fairness while at the same time pursue its core motivational goal of staying in high wellbeing. The final state of the emotion of joy, therefore, can be calculated as $E_{joy}(W, F) = f(W) \cdot g(F)$.
  On the contrary, the agent will be in a negative emotion of *sadness* when it is in a low wellbeing state (i.e., $W < 0$), because the situation is inconsistent with the agent's goal of pursuing individual benefits. The desirability of the emotion of $sadness$, $D_{sadness}$, is then equivalent to the value of $-W$. The intensity of emotion $sadness$ is then based on the increase or decrease of the agent's sense of social fairness, and the final state of the emotion of $sadness$ can be calculated as $E_{sadness}(W, F) = f(-W) \cdot g(F)$.

*2) **Different Strategies to Update Core/Secondary Derivation Functions**:* The core derivation function $f(D_x)$ monotonically maps the desirability of emotion x to a value in-between $[0, 1]$, with value 0 indicating no desirability of emotion x and value 1 indicating the highest desirability of emotion x. Function $f(D_x)$ can be formally defined as:

$$f(D_x) = D_x^{\mu}, \qquad 0 \leq D_x \leq 1, \mu > 0 \tag{10}$$

where μ is the *core derivation coefficient*. According to different values of μ, three different strategies to adapt the core derivation function can be defined as follows:

- **Conceder ($0 < \mu < 1$)**: The value of $f(D_x)$ increases quickly when the value of D_x is small and slowly when the value of D_x is high.
- **Linear ($\mu = 1$)**: The value of $f(D_x)$ increases at a constant rate with the value of D_x.
- **Boulware ($\mu > 1$)**: This strategy is the contrary of Conceder, which means that the value of $f(D_x)$ increases slowly when the value of D_x is small and quickly when the value of D_x is high.

The above three different strategies are originally used as time dependent tactics in agent negotiation to indicate different patterns of negotiation behaviors under time constrains [29]. These strategies are used here to indicate different kinds of agent behaviors as a response to the emotional stimulates. An agent who uses the Conceder strategy is more aggressive or sensitive because a slight desirability of an emotion will cause a dramatic increase in the corresponding value of emotion functions. In contrast, an agent who uses the Boulware strategy is more conservative or stubborn because only a high desirability of an emotion can cause a dramatic increase in the value of emotion functions. An agent who uses Linear strategy is moderate in nature because the increase rate of the emotion function is independent of the value of desirability.

Accordingly, the secondary derivation function $g(I_x)$ maps intensity $-1 \leq I_x \leq 1$ to a value in-between $[0, 1]$, which can

be defined by Equation 11.

$$g(I_x) = \left(\frac{I_x + 1}{2}\right)^{\nu} \qquad -1 \leq I_x \leq 1, \nu > 0 \qquad (11)$$

where ν is the *secondary derivation coefficient.*

C. The Interaction Protocol

Algorithm 1 gives the sketch of the interaction protocol for the emotional MARL framework. For a clear illustration, we exemplify by using Q-learning as the outer-layer learning algorithm and use $\pi(e)$ to denote the strategy of selecting emotion derivation function e (FW or WF) in the inner-layer learning. Strategy $\pi(e)$ can be updated using general RL approaches [7], such as TD control method Sarsa, Q-learning and Actor-Critic methods. For example, when using Q-learning, two Q values, $Q(FW)$ and $Q(WF)$, can be stored to indicate the future rewards of choosing function FW and WF, respectively, and based on these two values, agents can decide which function to choose to derive emotions.

Algorithm 1: The interaction protocol

```
1  Initialize network and learning parameters;
2  for each learning episode t (t=1,...,T) do
3      for each agent i (i=1,...,N) do
4          Chooses function e based on π(e);
5          if e=FW then
6              Chooses action a_i based on Q_i^FW(a);
7          else Chooses action a_i based on Q_i^WF(a);
8      for each agent i (i=1,...,N) do
9          for each neighbor j ∈ N(i) do
10             Plays action a_i with agent j;
11             Receives r_{i,j} after interacting with agent j;
12         Calculates its sense of social fairness;
13         Calculates its individual wellbeing;
14         Calculates R_int using selected function e;
15         Calls inner-layer learning to update π(e) using
           intrinsic emotional reward R_int;
16         Calls outer-layer learning to updates Q_i^e(a) using
           intrinsic emotional reward R_int.
```

In the emotional learning framework, all the agents interact with each other simultaneously and concurrently in each learning episode. As the focus of this paper is on adapting the definition of individual rewards in MARL to facilitate cooperation, we do not consider state-transitions when formulating the MARL algorithm, as did in [9]. Each agent keeps an inner-learning strategy $\pi(e)$ and Q values associated with function FW and function WF. In a learning episode, each agent chooses emotion derivation function e based on $\pi(e)$ (Line 4). If the selected function is function FW (Line 5), the agent then chooses best-response action a_i to cooperate or to defect based on $Q_i^{FW}(a)$ (Line 6), otherwise, action a_i is chosen based on $Q_i^{WF}(a)$ (Line 7). After choosing an action, each agent interacts with its neighbors simultaneously using action a_i, observes the actions of its neighbors, and receives the rewards from the environment after each interaction (Line 9-11). The agent then calculates its social fairness (Line 12) and individual wellbeing (Line 13) after each learning episode, and derives the intrinsic emotional reward R_{int} using the selected emotion derivation function e (Line 14). Finally, the agent calls the inner-layer learning to update strategy $\pi(e)$ of selecting emotion derivation functions (Line 15) and updates the learning information $Q_i^e(a)$ associated with function e using the intrinsic emotional reward R_{int} (Line 16).

TABLE II
PARAMETER SETTINGS IN THE EXPERIMENTS

Parameters in the experiments			
Categories	Parameters	Values	Meanings
Network	N	$[25, 1000]$	number of agents
	k	$\{2, 4, 6, 8, 12\}$	average neighborhood size in small-world networks
	ρ	$\{0, 0.2, 0.4, 0.6, 0.8, 1\}$	re-wiring probability in small-world networks
	m_0	5	initial agents to generate small-world networks
	m	1	number of edges added to the network at every step in small-world networks
	λ	3	exponent value of power law in scale-free networks
Learning algorithm	α	0.5	learning rate in Q-learning, WPL and WoLF-PHC
	α_l	0.1	learning rate when losing in WoLF-PHC
	α_w	0.04	learning rate when wining in WoLF-PHC
	β	0.5	learning rate to update aspiration level
	γ	0	discount factor
	ε	0.10	ε-exploration rate
Game structure	T	5	temptation to defect
	R	3	reward for mutual cooperation
	P	1	punishment for mutual defection
	S	0	sucker's pay-off
Others	h	10	scalable parameter in aspiration-based approach
	M	$\{1, 2, 3, 4, 5\}$	interaction steps in an episode
	μ	$\{0.2, 0.6, 1, 2, 3, 5, 10\}$	core derivation coefficient
	ν	$\{0.2, 0.6, 1, 2, 3, 5, 10\}$	secondary derivation coefficient
	p	$[0, 1]$	proportion of non-emotional leaners

V. EXPERIMENTS

This section presents the experimental studies. Subsection V-A introduces experimental settings, and Subsection V-B gives experimental results and analysis.

A. Experimental Settings

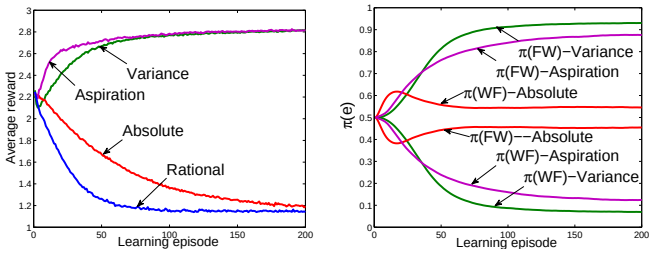
Table II lays out the parameter settings in the experiments. We use the typical values of the PD game pay-offs in Table I. The Watts-Strogatz model [30] is used to generate a small-world network, and the Barabasi-Albert model [18] is used to generate a scale-free network. To use the Barabasi-Albert model, we start with $m_0 = 5$ agents and add a new agent with $m = 1$ edge to the network at every time step. This network evolves into a scale-free network following a power law with an exponent $\lambda = 3$ [31]. Three different MARL algorithms are chosen for agent interactions. They are Q-learning [19] with ε-greedy exploration, Weighted Policy Learner (WPL) algorithm [32] and Win-or-Learn-Fast with Policy-Hill-Climbing (WoLF-PHC) [33]. The impacts of different learning algorithms on the learning performance will be tested. Discount factor γ is set to 0 as we do not consider the state transitions of agents. Two different learning rates are used in WoLF-PHC, based on whether an agent is winning or losing. Learning rate α_w is set to 0.04 when the agent is winning and learning rate α_l is set to 0.1 when the agent is losing. Unless stated otherwise, in all three learning approaches, learning rate α is set to 0.5, and ε is set to 0.1 in the ε-greedy exploration strategy to indicate a small probability of exploration.

The population and neighborhood size, and network randomness are important factors that can influence the emergence of cooperation in a system. We vary agent number N in the range of $[25, 1000]$ in network GR_N and network $SW_N^{4,0.4}$, neighborhood size k in the set of $\{2, 4, 6, 8, 12\}$ in network $SW_{100}^{k,0.4}$, and network randomness ρ in the set of $\{0, 0.2, 0.4, 0.6, 0.8, 1\}$ in network $SW_{100}^{4,\rho}$, to investigate the

impacts of these factors on the emergence of cooperation in the whole system. In the aspiration-based approach, scalable parameter h is set to 10 and learning rate β is set to 0.5. Unless stated otherwise, interaction steps M in each learning episode is set to 1 to model memoryless agents, the linear functions ($\mu = \nu = 1$) are used to model moderate agents in updating the core and secondary emotions, and Q-learning/WPL is used for the outer-layer/inner-layer learning algorithm. All results are averaged over 100 independent runs.

B. Experimental Results and Analysis

1) ***Emergence of Cooperation in The Emotional MARL Framework****:* We carried out extensive experiments applying the proposed emotional MARL framework in the spatial IPD. The patterns of results do not differ greatly in terms of population sizes (which can be seen later). Therefore, here, we only give the results when population size is 100.

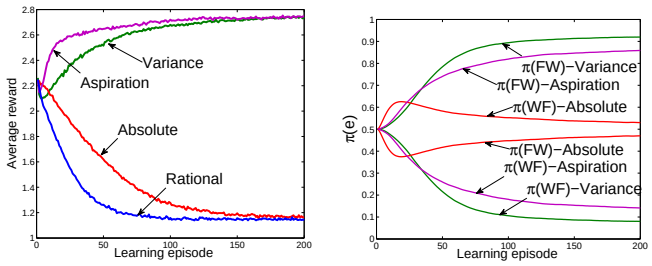


(a) Average reward (b) Function selection strategy

Fig. 3. Learning dynamics in a grid network.

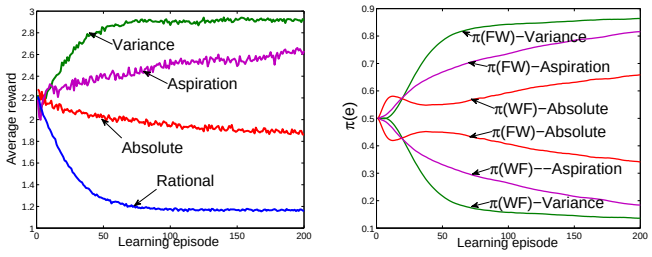
Fig. 3(a) shows the learning dynamics of the average population reward in grid network GR_{100}. In the figure, *Absolute value-based*, *Variance-based* and *Aspiration-based* learning approaches indicate the different approaches to appraise agents' individual wellbeing in the proposed emotional MARL framework, and *Rational* learning denotes the fully distributed MARL approach, in which agents make decisions by maximizing their Q-values directly based on extrinsic individual rewards from the environment. As can be seen, different learning approaches produce different patterns of learning behaviors. As expected, the rational learning approach ends up with mutual defection among the agents, causing a converged average population reward close to $P = 1$. This result indicates that behaving rationally does not necessarily bring about the best outcome for the learning agents. In the IPD game, rational learning can only lead to a behavior that ensures each agent from being exploited by its opponents. No agent can achieve a dominant position by choosing defection to exploit its opponent because the opponent will eliminate such dominance by also choosing defection, resulting in the Nash equilibrium of mutual defection among the agents. In contrast, the proposed emotional MARL framework using the variance-based or aspiration-based approach to appraise individual wellbeing can greatly facilitate the emergence of cooperation, causing a converged average population reward close to mutual cooperation of $R = 3$. These results indicate that the emotional learning framework can endow selfish agents with an internal cognitive and emotional capability that drives these agents to reciprocal behaviors in IPD. Learning with emotions can prevent agents from being Rational Fools [34], who act to maximize their gains in the short term. The absolute value-based approach, however, cannot bring about cooperation among the agents, causing a similar learning curve to the rational learning approach. This result confirms that the absolute wealth of an agent cannot reflect the agent's real emotion state regarding its own wellbeing.

Fig. 3(b) plots the dynamics of selecting emotion derivation functions in grid network GR_{100}, in which $\pi(e)$ indicates the probability of selecting emotion derivation function e (FW or WF) to derive the emotions of the agents. As can be seen, when agents adopt the satisfaction-based or the variance-based approach to appraise individual wellbeing, function FW gradually emerges as a dominant factor with a probability close to 100% for the agents to derive their emotions. When agents adopt the absolute value-based approach, however, neither of the functions can dominate the other. The probability of selecting function WF is only a bit higher than that of selecting function FW. The results in Fig. 3(b) indicate that, through the competition of agents' emotion derivation functions in the inner-layer learning, the socially reciprocal behavior using the FW function can override the selfish egoistic behavior using the WF function, which correspondingly facilitates cooperation among the agents. The results provide an explanation of real-life phenomena where people are social beings and often care about social fairness more than their own interests in order to achieve mutually beneficial outcomes [35]. For example, in Ultimatum Game, people usually refuse an unfair offer, even if this decision will cause them to receive nothing, and in Public Goods Game, people are usually willing to punish free riders, even though this punishment imposes a cost on themselves.



(a) Average reward (b) Function selection strategy

Fig. 4. Learning dynamics in a small-world network.



(a) Average reward (b) Function selection strategy

Fig. 5. Learning dynamics in a scale-free network.

Fig. 4 and Fig. 5 show learning dynamics in a small-world ($SW_{100}^{4,0.4}$) and a scale-free network ($SF_{100}^{k,3}$), respectively. We can see that the patterns of learning behaviors in these

two kinds of networks are almost the same as those in grid networks. The minor difference occurs in the scale-free network, in which the variance-based approach outperforms the aspiration-based approach in terms of a higher level of cooperation and a quicker emergence rate, and the absolute value-based approach can maintain a certain level of cooperation. In general, we can say that the proposed emotional-MARL framework enables different kinds of networks to emerge high levels of cooperation among the agents when variance-based and aspiration-based approaches are adopted in the appraisal of individual wellbeing.

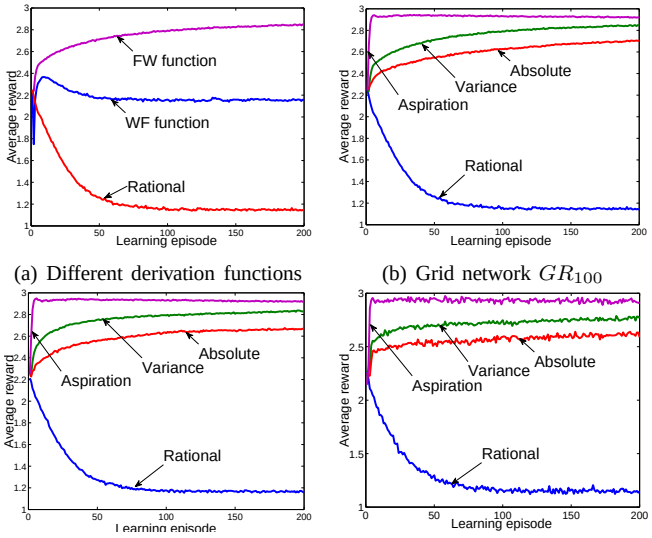
TABLE III
PERCENTAGE OF COOPERATION IN PD GAME (%)

The PD Game				
Approach		Grid	Small-world	Scale-free
Emotional learning	Variance-based	98.3 ± 0.3	99.1 ± 0.3	97.6 ± 0.7
	Aspiration-based	99.1 ± 0.2	99.2 ± 0.2	92.6 ± 1.3
	Absolute-based	11.2 ± 2.4	13.5 ± 3.2	36.3 ± 3.5
Rational learning		0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0

Table III summarizes the overall performance of the different learning approaches. The percentage of cooperation means the frequency of CC outcomes in 100 independent runs, and the final results (with their 90% confidence intervals) are averaged over 50 Monte Carlo simulations. To eliminate any probabilistic noise caused by the parameter settings, the learning rate α is randomly chosen in-between $[0, 1]$, and the exploration rate ε is randomly chosen in-between $(0, 0.5]$ to indicate a small probability of exploration. Each experimental configuration is kept the same for all the learning approaches in a run. From the results, we can see that in all networks, the rational learning approach results in a fully defective behavior of agents, while the variance-based and aspiration-based learning approaches can greatly facilitate the emergence of cooperation among agents. The cooperation levels using the variance-based and aspiration-based approaches are a bit lower than full cooperation of 100%. This is because when using a small learning rate α and a high exploration rate ε, there is a slight probability for the agents to converge to DD behavior.

2) ***Emergence of Cooperation Without Inner-layer Learning****:* Through the competition between function FW and WF in the inner-layer learning process, socially reciprocal behavior using function FW can always emerge as the dominant factor in deriving emotions in order to achieve cooperation among the agents. A question then arises whether agents can learn to achieve cooperation directly based on function FW without the inner-learning process.

Fig. 6(a) shows learning dynamics in grid network GR_{100} using only outer-layer learning. As expected, agents using function FW can learn to cooperate successfully, while agents using function WF can only learn to achieve a certain level of cooperation. This result can be explained that agents using function FW to derive their emotions are socially aware agents that behave in a reciprocal altruistic way, while agents using function WF to derive their emotions are selfish egoists that care about individual wellbeing more than social fairness. Figures 6(b)-(d) show learning dynamics in the three different kinds of networks when agents use function FW to derive emotions and use the proposed three different approaches to appraise individual wellbeing. As can be seen, three different



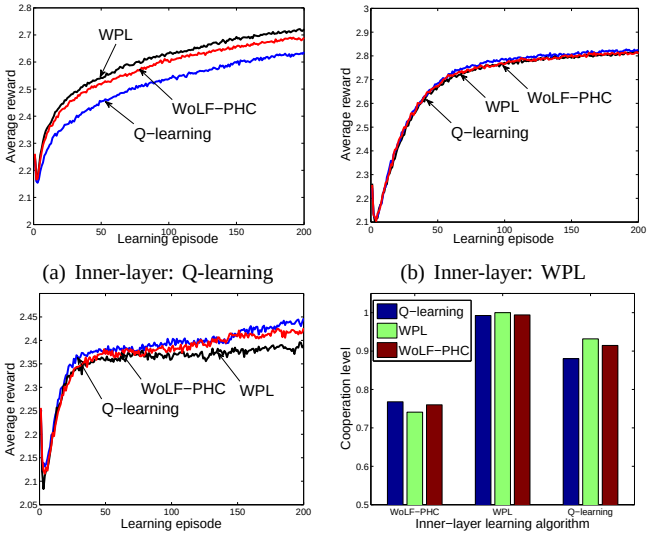
(a) Different derivation functions (b) Grid network GR_{100}

(c) Small-world network $SW_{100}^{4,0.4}$ (d) Scale-free network $SF_{100}^{k,3}$

Fig. 6. Learning dynamics using only outer-layer learning.

kinds of networks produce almost the same pattern of learning behaviors. The aspiration-based approach outperforms the other two approaches in all three networks. The absolute value-based approach can maintain a certain level of cooperation, which is a bit lower than that of using aspiration-based or variance-based approach. These results further confirm that absolute wealth cannot reflect agents' real wellbeing state in order to achieve high levels of cooperation among them.

3) ***Influence of MARL Algorithms****:* To test the influence of different MARL algorithms on the emergence of cooperation, agents use one of the three MARL algorithms for the inner-layer and outer-layer learning, respectively. Therefore, there are 9 different cases in total. Fig. 7(a)-(c) shows the learning dynamics when Q-learning, WPL and WoLF-PHC are used for inner-layer learning in grid network GR_{100}, respectively, while Fig. 7(d) shows the cooperation levels over the last 30 episodes using different MARL algorithms.



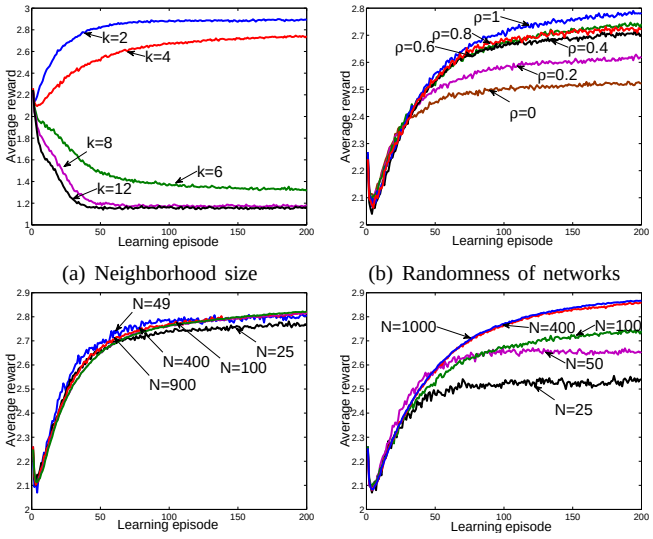
(a) Inner-layer: Q-learning (b) Inner-layer: WPL

(c) Inner-layer: WoLF-PHC (d) Cooperation levels

Fig. 7. Influence of different MARL algorithms.

From Fig. 7(a)-(c), we can see that the emotional learning framework can increase the levels of cooperation among the agents using different MARL algorithms for the outer-layer and inner-layer learning. Using WPL for inner-layer learning enables a more efficient emergence of cooperation in terms of a faster speed than that of using Q-learning, and a higher level of cooperation than that of using WoLP-PHC. Fig. 7(d) shows that WPL can achieve the highest levels of cooperation (close to 100%), followed by Q-learning, which can achieve a cooperation level of around 91%. WoLF-PHC achieves the lowest level of cooperation around 75%.

*4) **Influence of Neighborhood/population Size and Network Randomness**:* Fig. 8 shows the influences of different neighborhood sizes, orders of network randomness and population sizes on the emergence of cooperation when agents adopt the variance-based emotional learning approach.



(a) Neighborhood size (b) Randomness of networks

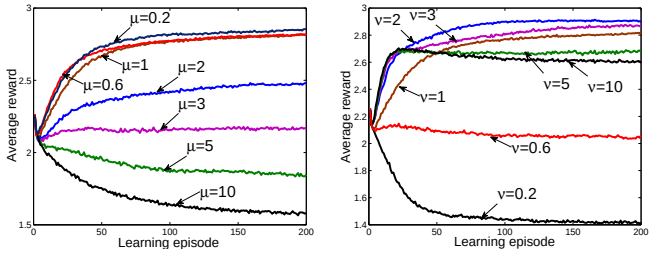
(c) Population size in GR_N (d) Population size in $SW_N^{4,0.4}$

Fig. 8. Learning dynamics with different neighborhood sizes, orders of network randomness and population sizes.

Fig. 8(a) shows the learning dynamics with different neighborhood sizes k in small-world network $SW_{100}^{k,0.4}$. We can see that only a small number of average neighbors ($k = 2, 4$) can ensure the emergence of cooperative behavior of the agents. When the number of average neighbors becomes large ($k = 6, 8, 12$), cooperation cannot be achieved. We carried out extensive experiments by varying the number of neighbors in different population sizes, and found that there was always a turning point of performance from cooperation to defection. This phenomenon is quite interesting and indicates that more local interactions of agents are not necessary to facilitate the emergence of an altruistic norm in social dilemma games, against the general cases of norm emergence in coordination games given by previous studies [31], [36]. Fig. 8(b) plots the learning dynamics with different orders of network randomness ρ in small-world network $SW_{100}^{4,\rho}$. The result shows that agents are able to achieve a high level of cooperation in a network with high randomness. This is because the increase in randomness can reduce the network diameter (i.e., the largest number of hops in order to traverse from one vertex to another), and the smaller a network diameter is, the more efficient for the network to evolve a social norm [36].

Fig. 8(c) and Fig. 8(d) show the learning dynamics with different population sizes in grid network GR_N and small-world network $SW_N^{4,0.4}$, respectively. We can see that in the grid network, the increase in population size has a negligible impact on the emergence of cooperation among the agents, while in the small-world network, the increase in population size can raise the cooperation level among the agents slightly.

*5) **Influence of Core/secondary Derivation Coefficients**:* The core/secondary derivation coefficients μ/ν are important factors that indicate different kinds of agent behaviors as a response to the emotional stimulates. Fig. 9 shows the impact of core/secondary derivation coefficient μ/ν on the emergence of cooperation in grid network GR_{100}.



(a) Core derivation coefficient μ (b) Secondary derivation coefficient ν

Fig. 9. Impact of derivation coefficients on the emergence of cooperation.

From Fig. 9(a), we can see that the Conceder strategies ($\mu = 0.2, 0.6$) and the Linear strategy ($\mu = 1$) to adapt the core derivation function can greatly boost the emergence of cooperation among the agents and these two strategies result in almost the same learning behaviors, while the Boulware strategies ($\mu = 2, 3, 5, 10$) hinder the establishment of cooperation. The larger the value of μ, the lower the level of cooperation among the agents. The result indicates that cooperation is more likely to emerge if agents are more aggressive in their core appraisal of emotions. Fig. 9(b) shows the impact of secondary derivation coefficient ν on the emergence of cooperation. The Conceder strategies ($\nu = 0.2, 0.6$) lead to low levels of cooperation, while the Linear strategy ($\mu = 1$) and the moderate Boulware strategies ($\nu = 2, 3$) can obtain high levels of cooperation. Drastic Boulware strategies ($\nu = 5, 10$), however, slightly decrease the levels of cooperation again. This result indicates that cooperation can be achieved if agents are conservative in their secondary appraisal of emotions. Fig. 9 shows that different agents' personalities (in terms of adapting the core and secondary derivation functions) have significant impacts on the emergence of cooperation among the agents.

*6) **Influence of memory size M**:* To test the influence of memory size M on the emergence of cooperation, we vary M in the set of $\{1, 2, 3, 4, 5\}$ in network GR_{100} so that agents update their emotional states in every M episodes.

Results in Fig. 10 show that cooperation emerges at almost the same speed when $M = 1$ and $M = 2$, but much more quickly when $M = 3$. Extending memory size further to $M = 4$ and $M = 5$, however, significantly hinders the emergence of cooperation. This result indicates that the frequency of updating intrinsic emotion states has a significant

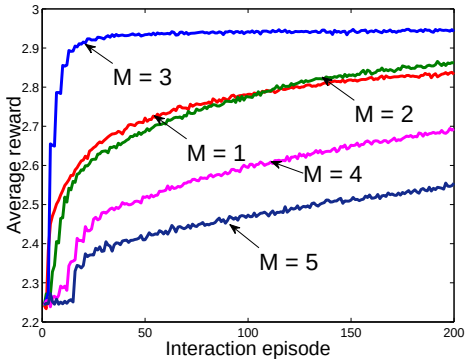
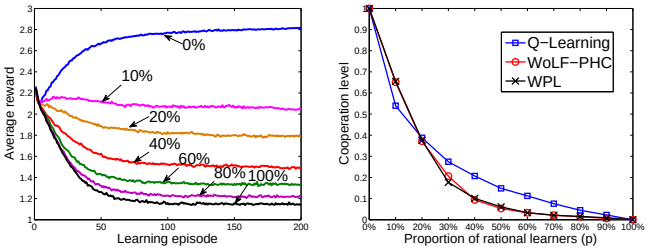


Fig. 10. Learning dynamics with different memory sizes (M).

impact on the emergence of cooperation among the agents. A higher updating frequency (short memory size) will cause a more dynamic extrinsic environment, which accordingly slows the emergence of cooperation, while a too low updating frequency (long memory size) will bring about a delay for the agents to catch up with the changing environment, causing a slow convergence of cooperation among the agents.

7) ***Influence of Agent Heterogeneities****:* To show the influence of agent heterogeneities on the emergence of cooperation, different proportions (p) of non-emotional (i.e., rational) learning agents are randomly deployed in a population of emotional learning agents in network GR_{100}. Fig. 11(a) shows the learning dynamics when emotional learning agents use WPL for inner-layer learning and Q-learning for outer-layer learning. Fig. 11(b) shows the levels of cooperation regarding to the proportion of rational learning agents when emotional learning agents use WPL for inner-layer learning.



(a) Learning dynamics (b) Levels of cooperation

Fig. 11. Impact of different proportions of rational learners.

Results in Fig. 11(a) show that when there is only a small proportion (e.g., $p = 10\%$) of rational agents in the population, the emergence of cooperation is significantly hindered, and further increasing the proportion of rational agents can steadily decrease the level of cooperation among agents. These results illustrate that even a small proportion of rational learners can greatly impede the emergence of cooperation among a large-scale society of emotional learning agents. Fig. 11(b) clearly shows that the three different outer-layer learning algorithms have the same pattern of results. When there are no rational learning agents in the population, all the agents can learn to cooperate with each other. When there are 10% rational learning agents, the cooperation levels decrease significantly to 65% when emotional learning agents use WoLF-PHC and WPL for outer-layer learning, and to 54% when emotional learning agents use Q-learning for outer-layer learning. The relationship between the decrease of cooperation level and the increase of proportion of rational learning agents is non-linear, but follows a logarithmical distribution.

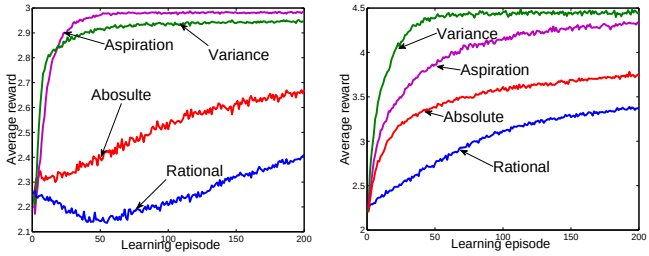
8) ***Solving Other Social Dilemmas****:* To show the generality of the proposed framework, we also apply it to solve other forms of social dilemmas in the Chicken Game (Table IV) and the Stag Hunt Game (Table V). Unlike the PD Game, both the Chicken Game and the Stag Hunt Game have two pure Nash equilibria. The two pure Nash equilibria (CD and DC) in the Chicken Game have the same social welfare, but they are not fair for the agent who chooses cooperation. In the Hunt Stag Game, however, the two pure Nash equilibria (CC and DD) are fair for both agents with CC dominating DD. It would be interesting to test whether the proposed framework is still effective for agents to learn cooperative behaviors in these social dilemmas.

TABLE IV
THE CHICKEN GAME

	C	D
C	3,3	1,5
D	5,1	0,0

TABLE V
THE STAG HUNT GAME

	C	D
C	5,5	0,1
D	1,0	3,3



(a) Chicken Game (b) Stag Hunt Game

Fig. 12. Solving other forms of social dilemmas.

Fig. 12 plots the learning dynamics using the four different approaches in grid network GR_{100}. In the Chicken game (Fig. 12(a)), the emotional learning approaches converge to average rewards that are much higher than the rational learning approach. This is because agents using the emotional learning approaches can learn a more deterministic CC behavior than the rational learning approach. When using the rational learning approach, agents can only achieve a certain level of cooperation, which means that there is a probability for the agents to converge to DD behavior, resulting in a much lower reward of 0. The same pattern of results can be observed in the Stag Hunt Game (Fig. 12(b)), in which the emotional learning approaches can guarantee higher levels of CC behavior than the rational learning approach.

Table VI summarizes the percentage of converging to CC behavior during independent 100 runs in 50 Monte Carlo simulations, in which the learning rate α is randomly chosen in-between $[0, 1]$, and the exploration rate ε is randomly chosen in-between $(0, 0.5]$ in each run. We can see that in the Chicken Game, the rational learning approach can only achieve cooperation levels that are slightly higher than 50%. This is because agents cannot distinguish between CD behavior and DC behavior using the rational learning approach. On the contrary, the emotional learning approaches can greatly

TABLE VI
PERCENTAGE OF COOPERATION (%)

The Chicken Game				
Approach		Grid	Small-world	Scale-free
Emotional learning	Variance-based	95.1 ± 1.4	95.1 ± 1.2	98.2 ± 0.6
	Aspiration-based	97.4 ± 1.2	96.3 ± 1.5	97.0 ± 1.0
	Absolute-based	63.8 ± 3.4	76.6 ± 4.2	73.1 ± 2.3
Rational learning		56.7 ± 5.4	58.2 ± 4.3	55.8 ± 4.6
The Stag Hunt Game				
Approach		Grid	Small-world	Scale-free
Emotional learning	Variance-based	96.5 ± 2.1	95.4 ± 1.2	92.3 ± 2.2
	Aspiration-based	92.4 ± 2.2	93.6 ± 2.7	86.6 ± 3.1
	Absolute-based	84.5 ± 4.7	76.1 ± 4.1	76.4 ± 3.4
Rational learning		76.6 ± 6.2	62.6 ± 5.2	66.4 ± 4.4

improve the cooperation levels because agents can consider the fairness among them so that the probability of converging to unfair behavior of CD or DC can be greatly decreased. In the Hunt Stag Game, however, the rational learning approach can achieve cooperation levels higher than 50%. This is due to the game structure in the Hunt Stag Game, in which the Nash equilibrium CC dominates the other Nash equilibrium DD. Even in this situation, the emotional learning approaches can further increase the cooperation levels among agents.

VI. RELATED WORK

This section reviews some of the most relevant studies in the current literature.

A. MARL to Solve Social Dilemmas

Numerous studies have investigated MARL in the context of social dilemmas. Sandholm and Crites [20] pioneered this area by conducting extensive experiments in which two Q-learning agents played IPD using lookup tables and simple recurrent neural networks to store the state-action estimates (Q-values). The length of history, the type of memory as well as the exploration mode were examined and the results showed that mutual cooperation did not occur when neither of the agents took their past actions into account and that the exploration mode had a major impact on the level of cooperation. Vrancx *et al.* [37] used a combination of replicator dynamics and switching dynamics to model multiagent learning automata in multi-state Prisoner's Dilemma (PD) games. Vassiliades and Christodoulou [23] proposed a method that evolved the payoffs of the IPD in order for RL agents to rapidly reach an outcome of mutual cooperation. Masuda *et al.* [21] analytically and numerically analyzed how an RL agent behaved against stochastic strategies and another learner in the IPD. Also, many other studies investigated MARL in social dilemmas based on the aspiration-based approaches [38], [39], [40], [41].

All these studies, however, focus on analyzing the learning dynamics between two agents and discovering the conditions under which naive RL agents can successfully learn to achieve cooperative behaviors. This focus makes all these studies different from ours because our work studies MARL of cooperation in spatial social dilemmas, where each agent's learning behavior is emotionally motivated.

In [9], Bazzan *et al.* studied MARL in spatial IPD using social attachments (i.e., belonging to a hierarchy or to a coalition) to lead learning agents in a grid to a certain level of cooperation. Although our work solves the same problem, we focus on exploiting emotions in modifying agent behavior during MARL, and do not impose assumptions of hierarchical supervision or coalition affiliation on the agents.

B. Emotions to Solve Social Dilemmas

Several studies have examined the evolution of cooperation in spatial Social Dilemmas by implementing an emotion mechanism. For example, Bazzan and Bordini [15] specified the rules of generating emotions based on the OCC model [42] by comparing an agent's received rewards to a pre-defined constant value and by counting the number of neighbors who are in a specific emotion state. The results showed that the level of cooperation in a two-dimensional IPD game when agents made decisions using emotions increased to 47%, a slightly higher than that of 32% without using emotions. Szolnoki *et al.* [16] proposed a new imitation mechanism that agents could copy the neighbors' emotional profiles (defined as the probability to cooperate with the neighbors of various success rates) and found that this imitation could be capable of guiding the population towards cooperation in three different social dilemmas. Bazzan *et al.* [34] used sentiments like generosity toward others and guilt for not having played fairly to prevent IPD players from trying to maximize the gain in the short term only. The results showed that in a society where agents had emotions, to behave rationally might not be the best attitude neither for an individual nor for the social group.

All these studies are based on rule-based emotional frameworks, in which the way of eliciting emotions must be predefined so that agents can choose their actions of cooperation or defection directly based on the emotion states. This is in contrast to our work, in which emotions are used as intrinsic rewards to bias agents' learning behaviors during repeated interactions with their neighbors in spatial social dilemmas.

C. Emotions for Agent Learning

Numerous studies have studied emotion-based agent learning. Ahn and Picard [11] presented a computational framework where both "the extrinsic reward from the external goal" and "the intrinsic reward from multiple emotion circuits and drives" played an integral role in agent learning. Their experimental results showed that using emotions could improve the speed of learning and regulate the trade-off between exploration and exploitation. Sequeira *et al.* [14] proposed four emotion appraisal dimensions (i.e., novelty, motivation, control and valence) to evaluate an agent relationship with its environment. Each of these dimensions was translated into a numerical feature that was used as intrinsic rewards to the agent in an RL context. The experimental results showed that contributions from different reward features could lead to distinct behaviors that allowed agents to adapt to particular environments and thus obtain better performance. Salichs and Malfaz [13] proposed a new approach to modeling emotions in agents based on drives, motivations and emotions. Three kinds of emotions (i.e., happiness, sadness and fear) were implemented to enable an agent to learn reasonable behaviors in order to maximize its wellbeing by satisfying its drives and avoiding dangerous situations.

Most of these studies, however, focus on single-agent scenarios, in which the emotion system drives a single agent to learn more efficient strategies and adaptive behaviors in complex environments. In our work, emotions are used to affect the behavior of multiple agents that conduct learning at the same time in order to achieve cooperation among these agents in the context of spatial social dilemmas. This focus differentiates our work from all these previous studies.

VII. Conclusion and Future Work

On one hand, it is important to incorporate emotion-related concepts and mechanisms into a computational model so as to make the model more aligned with human expectations and therefore more feasible to reach intended goals. On the other hand, achieving a satisfactory understanding of the emergence of cooperation in social dilemmas is also a fundamental issue for elucidating and comprehending many key problems that our societies are faced with today. This paper proposes an emotional MARL framework which endows agents with internal cognitive and emotional capabilities that can drive these agents to learn cooperative behaviors. Experimental results revealed that different ways of appraising emotions, various network topologies and agent heterogeneities had significant impacts on agent learning behaviors in the proposed framework, and under certain circumstances, high levels of cooperation could be achieved among the agents.

It is interesting to extend the framework proposed in this paper to solve other social dilemmas with higher complexities (e.g., continuous action of agents and N-player ($N > 2$) social dilemma games). We leave this issue for future work.

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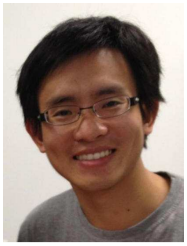
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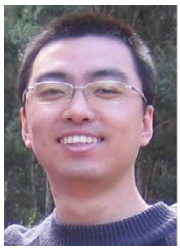
Chao Yu received the B.Sc. degree from the Huazhong University of Science and Technology, Wuhan, China, the M.Sc. degree from Huazhong Normal University, Wuhan, and the Ph.D. degree in computer science from the University of Wollongong, Wollongong, NSW, Australia, in 2007, 2010, and 2013, respectively.

He is currently a Lecturer with the School of Computer Science and Technology, Dalian University of Technology, Dalian, China. His current research interests include multiagent systems and learning, with their wide applications in modeling and solving various real-world problems in areas of Internet of Things and Vehicular Networking.



Minjie Zhang (SM'13) received the B.Sc. and M.Sc. degrees from Fudan University, Shanghai, China, and the Ph.D. degree in computer science from the University of New England, Armidale, NSW, Australia.

She is currently a Professor of Computer Science with the University of Wollongong, Wollongong, NSW, Australia. She is an active researcher and has published over 100 papers in the past ten years. Her current research interests include multiagent systems and agent-based modeling in complex domains.



Fenghui Ren received the B.Sc. degree from Xidian University, Xi'an, China, in 2003, and the M.Sc. and Ph.D. degrees from the University of Wollongong, Wollongong, NSW, Australia, in 2006 and 2010, respectively.

He is currently an Australia Research Council Discovery Early Career Researcher Award in Australia Fellow and a Lecturer with the School of Computer Science and Software Engineering, University of Wollongong. He is an active researcher and has published over 50 research papers. His current research interests include agent-based concept modeling of complex systems, data mining and pattern discovery in complex domains, agent-based learning, smart grid systems, and self-organizations in distributed and complex systems.



Guozhen Tan received the B.S. degree from the Shenyang University of Technology, Shenyang, China, the M.S. degree from the Harbin Institute of Technology, Harbin, China, and the Ph.D. degree from the Dalian University of Technology, Dalian, China.

He is currently a Professor and the Dean of the School of Computer Science and Technology, Dalian University of Technology. He was a Visiting Scholar at the Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign, Urbana, IL, USA, from 2007 to 2008. His current research interests include the Internet of things, cyber-physical systems vehicular Ad-hoc networks, intelligent transportation systems, and network optimization algorithms.

Prof. Tan was the recipient of the 2006 National Science and Technology Progress Award (second class) for his work in vehicle position and navigation, location-based service, traffic signal control, rapid respondence, and processing for traffic emergency. He was an Editor for the Journal of Chinese Computer Systems. He was a member of the China Computer Federation (CCF) and a committee man of the Internet Professional Committee of CCF, the Professional Committee of Software Engineering of CCF, and the Professional Committee of High Performance Computing of CCF.